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# N + - Network Fundamentals & Building Networks.

Introduction to WAN Technologies.

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# TASK 1.

## ❖ Create a comparison table listing at least 3 key differences between WAN and LAN, and give one real-world example of each from apps or services you use daily (for example, IRCTC, Instagram, or Zomato).

- **Differences Between WAN and LAN:**

| Basis | LAN (Local Area Network) | WAN (Wide Area Network) |
|---|---|---|
| 1. Coverage | Covers a small area, such as a home, office or college. | Covers large geographical areas, such as cities, countries or continents. |
| 2. Speed | Generally offers faster data transfer within the local network. | Speed depends on the network provider and connection. |
| 3. Ownership | Usually managed by an individual or organisation. | Often uses infrastructure managed by multiple service providers. |
| Real-World Example | A college LAN connects computers in different labs through switches. | IRCTC uses wide-area connectivity to serve users across India. |

# TASK 2.

❖ **Draw a simple network diagram using any drawing tool (or on paper and upload a photo) that shows how a user in Ahmedabad could connect to a Flipkart data center in Bangalore using a WAN, labeling the LAN and WAN segments clearly.**

➢ **WAN Connection Between Ahmedabad User and Flipkart Data Center in Bangalore:**

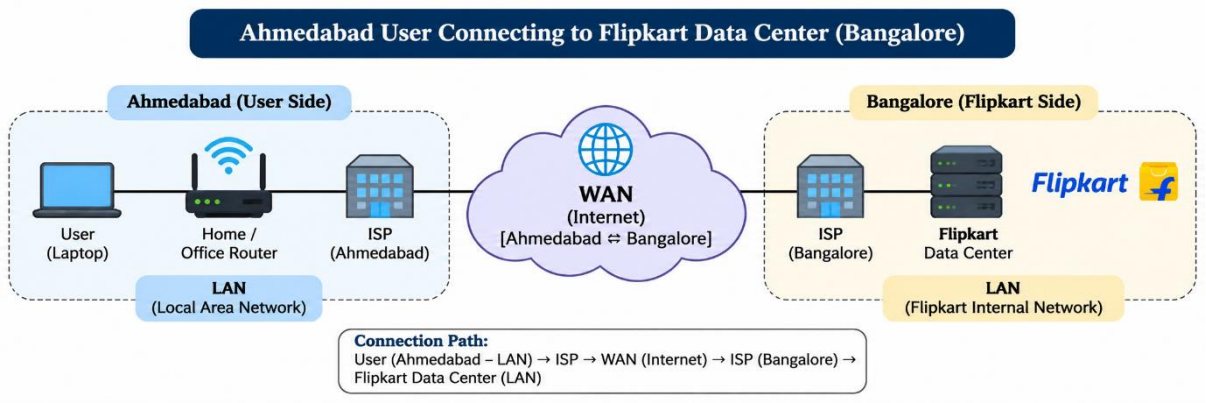



*Figure : WAN Connection Between Ahmedabad User and Flipkart Data Center in Bangalore.*

**Explanation:** The user's laptop in Ahmedabad connects to the local router and ISP through a LAN. The data then travels through the WAN (Internet) to Flipkart's data center in Bangalore, where the request is processed.

**Connection Path:** Ahmedabad User → Home Router → Ahmedabad ISP → WAN (Internet) → Bangalore ISP → Flipkart Data Center.

# TASK 3.

## ❖ Research and write a short summary (5-7 lines) explaining what MPLS is and how it improves data routing compared to traditional IP routing.

- MPLS (Multiprotocol Label Switching) is a technology used to transfer data efficiently across a network. It assigns short labels to data packets instead of checking the full IP address at every router. Routers use these labels to forward packets along predefined paths. This makes data forwarding faster and more efficient. Compared to traditional IP routing, MPLS can provide better traffic management, reliability, and Quality of Service (QoS). It is commonly used by service providers and large organisations to manage network traffic.

# TASK 4.

## ❖ Observe the pre-configured MPLS network in the provided Packet Tracer or GNS3 file, and note down two routing behaviors that are different from standard IP routing, Look for how labels are used and how packets move between routers.

- **Two Routing Behaviours of MPLS:**

  **Label Based Forwarding:** MPLS routers use short labels to forward packets instead of checking the full destination IP address at every router.

  **Predefined Paths:** Packets follow Label Switched Paths (LSPs) through the network rather than making an independent routing decision at every router.

  **Result:** MPLS provides more controlled and efficient packet forwarding compared to traditional IP routing.

# TASK 5.

## ❖ List three specific benefits MPLS offers over traditional routing for companies running high-traffic apps like YouTube or BookMyShow, and briefly explain how each benefit helps in real-world scenarios.

- **Three Benefits of MPLS Over Traditional Routing:**

  **Faster Data Forwarding:** MPLS uses labels to forward packets efficiently. For example, YouTube users can experience smoother video streaming with less delay.

  **Better Traffic Management:** MPLS can prioritise important traffic. For example, BookMyShow can give priority to booking transactions during high-demand ticket sales.

  **Improved Reliability:** MPLS supports predefined paths and alternative routes. If one path fails, traffic can be redirected to another path, helping services remain available.